

Discrete Geodesic Facet Vectorization for Conservative Thermodynamic Transport on Planetary Spheres: The Geometric Primitive $\vec{\mathbf{L}}_{AB} = \mathbf{v}_B - \mathbf{v}_A$

Web of Life Systems Architecture & Research Team

Planetary Thermodynamic Monad Modeling Group

<https://github.com/pascalranoroarijaona/WebOfLife>

September 15, 2026

Abstract

Planetary-scale biogeochemical and thermodynamic simulations require robust discrete spatial representations that eliminate polar coordinate singularities and support strictly conservative lateral fluxes. The Web of Life simulation architecture employs a Discrete Global Grid System (DGGS) based on hierarchical H3 hexagonal partitions on the spherical geoid $\mathbb{S}^2$. In this work, we present the design, mathematical formulation, and thermodynamic validation of the unnormalized edge displacement primitive $\vec{\mathbf{L}}_{AB} = \mathbf{v}_B - \mathbf{v}_A$, implemented via `computeBoundarySegmentVector3D`. We demonstrate that retaining the unnormalized 3D Cartesian vector between spherical boundary vertices preserves fundamental topological and physical invariants: exact anti-symmetry ($\vec{\mathbf{L}}_{BA} = -\vec{\mathbf{L}}_{AB}$), closed Stokesian contour circulation ($\sum_{k=1}^N \vec{\mathbf{L}}_{k,k+1} = \mathbf{0}$), and strict adherence to the First and Second Laws of Thermodynamics during interfacial advection and Fickian-Fourier diffusion. By decoupling boundary metrics from premature trigonometric projections, our approach guarantees machine-epsilon energy-mass conservation and unconditionally non-negative entropy generation ($\dot{S}_{\text{gen}} \geq 0$) across continuous planetary transport networks.

1 Introduction

Spatial discretization of planetary surfaces for computational fluid dynamics, climate dynamics, and whole-earth biogeochemical modeling has progressively transitioned from latitude-longitude rectangular rasters toward Discrete Global Grid Systems (DGGS) [1]. Standard equi-angular coordinates introduce severe polar singularities, metric distortion, and localized Courant-Friedrichs-Lewy (CFL) time-step bottlenecks. Hexagonal geodesic discretizations, particularly the H3 hierarchical aperture-7 spatial index, offer near-uniform cell areas, isotropic neighborhood connectivity, and reduced quantization error [2].

In the *Web of Life* planetary simulation engine (<https://github.com/pascalranoroarijaona/WebOfLife>), hexagonal cells represent thermodynamic monads—autonomous finite-volume containers maintaining state vectors of internal energy (U), sensible enthalpy, water vapor ($Q_{\text{H}_2\text{O}}$), dissolved inorganic carbon (DIC), oxygen, and mineral macronutrients [3]. Lateral transport between adjacent monads occurs across shared boundary facets. A ubiquitous failure mode in spherical finite-volume solvers is the premature normalization or spherical-polar projection of boundary edge vectors. Normalizing displacement vectors prior to computing directed facet areas introduces rounding errors, complicates Euclidean chord length calculation, and impairs exact metric cancellation across shared topological interfaces.

This paper details Sprint 061, which introduces and verifies the fundamental geometric primitive `computeBoundarySegmentVector3D` in the Earth-Centered, Earth-Fixed (ECEF) Cartesian

coordinate space $\mathbb{R}^3$. We formalize the integration of this vector primitive into interfacial finite-volume equations, demonstrating that strict algebraic anti-symmetry directly yields First Law mass-energy conservation and Second Law entropy dissipation compliance.

2 Mathematical and Geometric Formulation

2.1 Coordinate Space Parameterization

Let the planetary surface be embedded in 3D Cartesian space with the coordinate origin situated at the planetary center of mass. A boundary vertex $\mathbf{v}_A$ on a sphere of radius $R_\oplus = 6,371,008.8$ m is represented as:

$$\mathbf{v}_A = \begin{bmatrix} x_A \\ y_A \\ z_A \end{bmatrix} \in \mathbb{R}^3, \quad \|\mathbf{v}_A\|_2 = R_\oplus \quad (1)$$

For two adjacent vertices A and B defining an edge of a spherical H3 polygon Ω_i , the unnormalized boundary segment displacement vector $\vec{\mathbf{L}}_{AB}$ is defined as:

$$\vec{\mathbf{L}}_{AB} = \mathbf{v}_B - \mathbf{v}_A = \begin{bmatrix} x_B - x_A \\ y_B - y_A \\ z_B - z_A \end{bmatrix} \quad (2)$$

2.2 Topological and Metric Invariants

Retaining the unnormalized Cartesian difference vector guarantees three structural invariants:

1. Skew-Symmetry / Interfacial Parity:

$$\vec{\mathbf{L}}_{BA} = \mathbf{v}_A - \mathbf{v}_B = -(\mathbf{v}_B - \mathbf{v}_A) = -\vec{\mathbf{L}}_{AB} \quad (3)$$

When cell Ω_i evaluates edge (A, B) and neighbor Ω_j evaluates edge (B, A) , the underlying geometric vectors are exactly negated down to IEEE-754 binary64 precision without numerical drift.

2. Euclidean Chord and Spherical Arc Metrization: The straight-line chord length l_{chord} is recovered directly from the Euclidean L_2 norm:

$$l_{\text{chord}} = \|\vec{\mathbf{L}}_{AB}\|_2 = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2} \quad (4)$$

The exact geodesic great-circle arc length s_{AB} is then:

$$\theta = 2 \arcsin \left(\frac{\|\vec{\mathbf{L}}_{AB}\|_2}{2R_\oplus} \right), \quad s_{AB} = R_\oplus \theta \quad (5)$$

3. Discrete Stokes Loop Closure: For any closed N -sided spherical boundary contour $\partial\Omega_i = \bigcup_{k=1}^N \mathbf{e}_{k,k+1}$ (where $\mathbf{v}_{N+1} \equiv \mathbf{v}_1$), the contour integral of displacement vanishes identically:

$$\oint_{\partial\Omega_i} d\vec{\mathbf{l}} = \sum_{k=1}^N \vec{\mathbf{L}}_{k,k+1} = \sum_{k=1}^N (\mathbf{v}_{k+1} - \mathbf{v}_k) \equiv \mathbf{0} \quad (6)$$

2.3 Oriented Facet Normal and Area Vector

To calculate fluxes perpendicular to the boundary facet within the spherical tangent plane, the edge midpoint radial unit vector $\hat{\mathbf{r}}_{\text{edge}}$ is constructed:

$$\mathbf{v}_{\text{mid}} = \frac{1}{2}(\mathbf{v}_A + \mathbf{v}_B), \quad \hat{\mathbf{r}}_{\text{edge}} = \frac{\mathbf{v}_{\text{mid}}}{\|\mathbf{v}_{\text{mid}}\|_2} \quad (7)$$

The oriented tangent-plane normal vector $\vec{\mathbf{n}}_{\text{facet}}$ is obtained via the Cartesian cross product with the unnormalized segment vector:

$$\vec{\mathbf{n}}_{\text{facet}} = \hat{\mathbf{r}}_{\text{edge}} \times \vec{\mathbf{L}}_{AB} \quad (8)$$

Because $\hat{\mathbf{r}}_{\text{edge}} \perp \vec{\mathbf{L}}_{AB}$ is approximately satisfied for small cell diameters ($l_{\text{chord}} \ll R_{\oplus}$), the magnitude $\|\vec{\mathbf{n}}_{\text{facet}}\|_2 \approx \|\vec{\mathbf{L}}_{AB}\|_2$. For an atmospheric or oceanic fluid column of vertical thickness H_e , the directed boundary area vector $\mathbf{A}_{AB}$ is:

$$\mathbf{A}_{AB} = \left(\hat{\mathbf{r}}_{\text{edge}} \times \vec{\mathbf{L}}_{AB} \right) \frac{s_{AB}}{l_{\text{chord}}} H_e \quad (9)$$

3 Thermodynamic Transport Formulation

3.1 First Law Conservative Interfacial Balances

Let cell monad Ω_i contain internal energy U_i and chemical species masses $M_{i,k}$ for $k \in \{\text{water, carbon, oxygen, minerals}\}$. The finite-volume continuity equation governing the cell state is:

$$\frac{dM_{i,k}}{dt} = - \sum_{j \in \mathcal{N}(i)} \mathcal{F}_{ij,k} + \mathcal{S}_{i,k} \quad (10)$$

where $\mathcal{F}_{ij,k}$ is the net interfacial mass flux from cell i to cell j across facet AB , and $\mathcal{S}_{i,k}$ denotes internal metabolic or chemical sources.

Given a horizontal advection velocity field $\mathbf{u} \in \mathbb{R}^3$ evaluated at the boundary facet, the volumetric advective flux $\dot{V}_{AB}$ is:

$$\dot{V}_{AB} = \mathbf{u} \cdot \mathbf{A}_{AB} = \mathbf{u} \cdot \left(\hat{\mathbf{r}}_{\text{edge}} \times \vec{\mathbf{L}}_{AB} \right) \frac{s_{AB}}{l_{\text{chord}}} H_e \quad [\text{m}^3 \cdot \text{s}^{-1}] \quad (11)$$

Due to the skew-symmetry of the cross product and segment vector:

$$\mathbf{A}_{BA} = \hat{\mathbf{r}}_{\text{edge}} \times \vec{\mathbf{L}}_{BA} = \hat{\mathbf{r}}_{\text{edge}} \times (-\vec{\mathbf{L}}_{AB}) = -\mathbf{A}_{AB} \quad (12)$$

Consequently, the interfacial advective flux satisfies exact anti-symmetry:

$$\dot{V}_{BA} \equiv -\dot{V}_{AB} \implies \mathcal{F}_{ji,k} = -\mathcal{F}_{ij,k} \quad (13)$$

Summing over all pairwise cell interfaces on the closed spherical manifold guarantees strict global conservation:

$$\sum_i \sum_{j \in \mathcal{N}(i)} \mathcal{F}_{ij,k} \equiv 0 \quad (14)$$

3.2 Second Law Entropy Generation in Diffusive Transport

Interfacial thermal conduction and Fickian mass diffusion between adjacent cell centroids $\mathbf{x}_i$ and $\mathbf{x}_j$ separated by geodesic distance d_{ij} are formulated as:

$$\dot{Q}_{\text{cond}} = -\kappa_{\text{th}} \left(\frac{T_j - T_i}{d_{ij}} \right) \|\mathbf{A}_{AB}\|_2 \quad [\text{W}] \quad (15)$$

$$\dot{M}_{k,\text{diff}} = -D_k \left(\frac{c_{j,k} - c_{i,k}}{d_{ij}} \right) \|\mathbf{A}_{AB}\|_2 \quad [\text{kg} \cdot \text{s}^{-1}] \quad (16)$$

The local entropy generation rate $\dot{S}_{\text{gen},AB}$ associated with thermal conduction across facet AB is given by:

$$\dot{S}_{\text{gen},AB} = \dot{Q}_{\text{cond}} \left(\frac{1}{T_j} - \frac{1}{T_i} \right) = \kappa_{\text{th}} \frac{(T_i - T_j)^2}{T_i T_j d_{ij}} \|\mathbf{A}_{AB}\|_2 \geq 0 \quad (17)$$

Because $\|\mathbf{A}_{AB}\|_2 > 0$ and $T_i, T_j > 0$, the production of entropy is strictly non-negative, fulfilling the Clausius-Duhem inequality and the Second Law of Thermodynamics.

4 Implementation and Computational Verification

4.1 Algorithmic Realization

The primitive `computeBoundarySegmentVector3D` is implemented in TypeScript within `src/spatial/h3_adjacent`. The function accepts two 3D Cartesian coordinates conforming to the immutable `Vector3D` interface:

$$\text{computeBoundarySegmentVector3D}(v_1, v_2) = (v_2.x - v_1.x, v_2.y - v_1.y, v_2.z - v_1.z) \quad (18)$$

Input validation enforces strict finiteness via `Number.isFinite`, immediately throwing an exception upon detecting `NaN` or `Infinity`. For coincident coordinates ($v_1 = v_2$), the function deterministically yields the null vector $(0, 0, 0)$ without division-by-zero or loss of precision.

4.2 Empirical Verification Results

Validation suites implemented in `tests/sprint_061.test.ts` confirm numerical stability and physical invariants:

Test Category	Target Invariant	Observed Metric	Status
Orthogonal Basis	$\vec{\mathbf{L}}_{AB} = \mathbf{e}_y - \mathbf{e}_x$	$(-1.0, 1.0, 0.0)$	Passed
Antisymmetry	$\vec{\mathbf{L}}_{AB} + \vec{\mathbf{L}}_{BA} = \mathbf{0}$	$\ \vec{\mathbf{L}}_{AB} + \vec{\mathbf{L}}_{BA}\ < 10^{-16}$	Passed
Stokes Closed Loop	$\sum_{k=1}^6 \vec{\mathbf{L}}_{k,k+1} = \mathbf{0}$	$\ \sum \vec{\mathbf{L}}\ < 2.2 \times 10^{-15}$	Passed
First Law Conservation	$\Delta M_i + \Delta M_j = 0$	Exact 0.0 kg	Passed
Second Law Entropy	$\dot{S}_{\text{gen}} \geq 0$	$\min(\dot{S}_{\text{gen}}) \geq 0.0 \text{ J} \cdot \text{K}^{-1}$	Passed

Table 1: Numerical verification benchmarks for `computeBoundarySegmentVector3D` on H3 resolution 4 boundary sets.

5 Conclusion and Next Steps

Sprint 061 establishes the geometric foundation for flux vectorization in the Web of Life planetary engine. By formulating and deploying `computeBoundarySegmentVector3D`, boundary segment displacements retain unnormalized Cartesian accuracy, guaranteeing skew-symmetry, closed-loop Stokesian circulation closure, and First/Second Law thermodynamic consistency. Subsequent sprints will compose this primitive into comprehensive finite-volume interfacial solvers for planetary moisture circulation, thermohaline oceanic transport, and global carbon sequestration dynamics.

Code Availability

The full source code, test suites, and integration benchmarks are available under an open-source license at:

<https://github.com/pascalranoroarijaona/WebOfLife>

References

- [1] K. Sahr, D. White, and A. J. Kimerling, “Geodesic discrete global grid systems,” *Cartography and Geographic Information Science*, vol. 30, no. 2, pp. 121–134, 2003.
- [2] I. Brodsky, “H3: A hexagonal hierarchical spatial index,” *Uber Engineering Blog*, 2018.
- [3] P. Ranoroarijaona, “Web of Life: A Thermodynamic, Multi-Layer Monad Simulation of Biospheric and Biogeochemical Homeostasis,” *Web of Life Research Technical Reports*, 2024. <https://github.com/pascalranoroarijaona/WebOfLife>.